



Applying the SLR & Mapping Methodology

A systematic literature review (SLR) combined with mapping methodology provides the evidence-based foundation for designing effective caregiver support tools. This rigorous approach transforms research into actionable app features, ensuring your solution addresses real needs with proven strategies.

The methodology guides you from defining precise research questions through to extracting design requirements. Each step builds upon the previous, creating a traceable path from academic literature to practical application features that genuinely support informal caregivers.

Step 1: Define Research Questions

*Your research questions guide the entire process. You need both **Mapping Questions (MQ)** to understand the landscape and **Systematic Review Questions (RQ)** to find specific solutions. This is the most critical step.*

Mapping Questions

The "What exists?" questions:

- *What is the prevalence and nature of ethical dilemmas faced by informal caregivers?*
- *What are the primary categories of these dilemmas?*
- *What existing technological solutions have been developed?*
- *Which disciplines are most active in this research area?*

Systematic Review Questions

The "How is it solved?" questions:

- *What decision-making frameworks are proven effective for caregivers navigating ethical dilemmas?*
- *What are the key design requirements and usability considerations for technology-based interventions?*

Step 2: Define PICOC

PICOC defines the scope of your systematic literature review with precision, ensuring you capture relevant research whilst filtering out irrelevant studies. This framework provides clarity and consistency throughout the review process.

	Population <i>Informal caregivers (family, friends) of individuals with dementia</i>
	Intervention <i>Strategies, frameworks, guidelines, or technological tools for managing ethical dilemmas</i>
	Comparison <i>Not applicable for a foundational review, or could be "care as usual" without formal support</i>
	Outcomes <i>Identified ethical dilemmas, effective support strategies, design requirements for apps</i>
	Context <i>Community-based dementia care (home settings)</i>

Step 3: Define Inclusion & Exclusion Criteria

These criteria act as your filters during the paper selection process, ensuring consistency and rigour. They transform subjective decisions into objective, reproducible choices that strengthen the credibility of your review.

Inclusion Criteria

- *The paper discusses ethical dilemmas faced by informal caregivers*
- *The paper proposes or evaluates a support strategy, framework, or technological tool*
- *The primary context is dementia care*
- *The paper is a peer-reviewed journal article, conference proceeding, or systematic review*
- *The paper is written in English*

Exclusion Criteria

- *The paper focuses solely on formal/professional caregivers (e.g., nurses, doctors)*
- *The paper discusses ethical dilemmas in a clinical setting without relevance to the home environment*
- *The paper is a thesis, book chapter, or opinion piece without empirical data*
- *The full text is not accessible*

Step 4: Select Databases & Formulate Queries

Cast a wide net across multiple disciplines to ensure comprehensive coverage. Different databases index different journals and conferences, so searching multiple sources is essential for a thorough review.



Healthcare/Bioethics

PubMed, PsycINFO, CINAHL, Cochrane Library



Computer Science/HCI

ACM Digital Library, IEEEExplore, Scopus, Web of Science



Multidisciplinary

Google Scholar for grey literature and cross-discipline discovery

Example Query for PubMed

```
("informal caregiver*" OR "family caregiver*") AND (dementia OR alzheimer*) AND ("ethical dilemma*" OR moral distress OR decision-making) AND (support OR intervention OR framework OR "mobile app" OR technology)
```

You must adapt this query using the specific syntax of each database.



Step 5: Plan the Review Phases

This is your workflow for processing papers systematically. Each phase narrows the pool of studies, moving from broad discovery to focused analysis. Documenting numbers at each stage demonstrates thoroughness and transparency.



Identification

Run the query on all selected databases

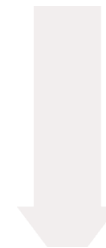
Result: e.g., n=1,200 records



Screening (Title/Abstract)

Remove duplicates and apply inclusion/exclusion criteria to titles and abstracts

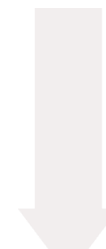
Result: e.g., n=150 records



Eligibility (Full Text)

Obtain and read the full text of remaining papers, applying criteria rigorously

Result: e.g., n=45 records



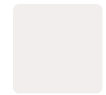
Inclusion (Snowballing)

Check reference lists of final papers for other relevant studies you might have missed

Result: Final corpus for analysis

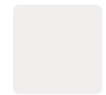
Step 6: Quality Assessment

Create a checklist to evaluate the trustworthiness of each paper. Quality assessment ensures that your app's foundation rests on robust, credible research rather than weak or biased studies.



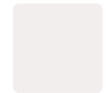
Methodology Clarity

Is the research methodology clearly described?



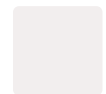
Empirical Foundation

Is the study based on empirical data (qualitative or quantitative)?



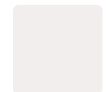
Participant Description

Are the study participants (caregivers) well-described?



Limitations Discussed

Are the limitations of the study discussed?



Design Process Detail

For design papers: Is the design/evaluation process clearly detailed?

Step 7: Ensure Traceability

Document every single step. This is your audit trail and is crucial for the credibility of your app's foundation. Traceability demonstrates rigour and allows others to verify or replicate your methodology.



Use a spreadsheet to log:

- *Paper ID, Author, Title, Year, Source*
- *Database where it was found*
- *Reason for inclusion/exclusion at the abstract stage*
- *Reason for inclusion/exclusion at the full-text stage*
- *Notes on key findings relevant to your research questions*

This systematic documentation transforms your review from a subjective exercise into a transparent, reproducible scientific process.

From SLR to App Design: The Direct Link

Your systematic literature review findings become the direct input for your app's features and content. This evidence-based approach significantly increases your app's chances of being useful, used, and effective for informal caregivers.

<i>SLR Finding</i>	<i>Proposed App Feature</i>
<i>Common Dilemma: Balancing truth-telling with preventing distress</i>	<i>Interactive decision-tree for "Should I tell the truth?" with pros/cons and suggested scripts</i>
<i>Effective Strategy: The 4-quadrant model (Medical Indications, Patient Preferences, Quality of Life, Contextual Features)</i>	<i>A guided form based on the 4 quadrants to help structure the user's thinking about a dilemma</i>
<i>Design Requirement: Use simple, non-clinical language</i>	<i>All app copy is written at an 8th-grade reading level and avoids medical jargon</i>
<i>Identified Gap: Lack of immediate, accessible support</i>	<i>A "Crisis Compass" section with quick-access guidelines for common, high-stress dilemmas</i>

By following this methodology, you move from an idea to an evidence-based design, creating a tool that genuinely addresses the needs of informal caregivers navigating complex ethical challenges in dementia care.